

Agronomic Guide for Seven Crops

Maize | Sorghum | Pearl Millet | Cowpeas | Groundnuts | Sunflowers | Cotton

Prepared for Zimbabwean summer-rainfall production conditions. Fertiliser, lime and pesticide decisions should be confirmed through soil testing, current product labels and advice from AGRITEX or a qualified agronomist.

Note: Values are practical guide ranges rather than fixed prescriptions. Variety, Natural Region, rainfall distribution, field history and soil-test results can alter recommendations.

MAIZE

Planting period

Plant with the first effective rains, normally mid-November to mid-December in much of Zimbabwe. Earlier planting is possible in cooler or higher-rainfall areas where moisture is reliable.

Planting conditions and land preparation

Use a well-prepared, weed-free, well-drained seedbed with adequate soil moisture. Plough or rip before the rains where possible; use conservation agriculture, mulch and planting basins where appropriate.

Growth stages

Germination, vegetative growth, tasselling, silking, grain filling and physiological maturity.

Planting method, spacing and depth

Plant in rows or basins. Typical spacing is 75-90 cm between rows and 20-30 cm within rows. Place seed about 5 cm deep; plant deeper only in dry sandy soil.

Fertiliser and weeding

Apply basal fertiliser according to a soil test, commonly Compound D/L at planting. Apply nitrogen top dressing at about 3-6 weeks after emergence, often split where rainfall is uncertain. Keep weed-free especially for the first 4-6 weeks.

Water and drought

Usually needs about 500-800 mm of well-distributed seasonal rainfall. Moisture is most critical around tasselling, silking and grain filling. Use early-maturing/drought-tolerant seed, early planting, mulch and moisture-conserving tillage in dry areas.

Pests and diseases

Key pests: fall armyworm, stalk borers, cutworms, aphids and storage weevils. Key diseases: maize streak virus, grey leaf spot, leaf blight, rust and smut. Streak causes pale yellow streaks and stunting; grey leaf spot makes long rectangular lesions; smut forms swollen dark galls.

Disease management

Use certified resistant seed, rotate crops, destroy heavily infected residues, plant early, maintain balanced nutrition and manage insect vectors. Use registered fungicides only where justified and according to label directions.

Harvest and post-harvest

Harvest when cobs are mature and grain is hard. Dry grain to about 12-13 percent moisture, shell carefully, remove damaged grain and store in clean, dry hermetic bags or pest-protected stores.

Soil requirements

Best on fertile, deep, well-drained loam or clay-loam soils, ideally pH 5.5-7.0. Avoid waterlogging and correct strong acidity with lime after a soil test.

SORGHUM

Planting period

Plant with effective rains, generally mid-November to mid-December. It can be a better late option than maize in lower-rainfall areas.

Planting conditions and land preparation

Prepare a fine, firm and weed-free seedbed because seed is small. Use well-drained fields and avoid sites heavily infested with Striga (witchweed).

Growth stages

Germination, vegetative growth/tillering, booting, heading, flowering, grain filling and maturity.

Planting method, spacing and depth

Row planting gives easier weeding. Use roughly 60-75 cm rows and 10-15 cm between plants after thinning. Sow shallowly, about 2-5 cm deep.

Fertiliser and weeding

Use fertiliser guided by a soil test; basal fertiliser is applied at planting and modest nitrogen may be top-dressed around 4-6 weeks after emergence. Weed well during the first 4-6 weeks and manage Striga through rotation and resistant varieties.

Water and drought

Usually performs with about 400-600 mm rainfall and tolerates dry spells better than maize. Flowering and grain filling remain sensitive periods.

Pests and diseases

Important pests include shoot fly, stem borers, aphids, midge, head bugs and Quelea birds. Important diseases include leaf blight, ergot, downy mildew, anthracnose and grain mould. Ergot produces sticky honey-like material from florets; grain mould discolours grain.

Disease management

Use tolerant varieties, clean seed, crop rotation, early planting and timely harvest. Avoid prolonged wet weather at maturity where possible and control volunteer hosts.

Harvest and post-harvest

Cut mature heads when grain is hard, dry thoroughly, thresh and clean. Store grain at about 12-13 percent moisture in dry, insect-protected storage.

Soil requirements

Adapted to many soils, including sandy loams and clay loams, provided drainage is good. It generally tolerates pH about 4.5-7.5 better than maize.

PEARL MILLET

Planting period

Plant with the first effective rains, commonly mid-November through December. It is especially suitable in the driest production zones.

Planting conditions and land preparation

Make a fine, firm seedbed and plant shallowly. Avoid waterlogged fields; conservation agriculture can help retain moisture.

Growth stages

Germination, vegetative growth, tillering, booting, heading, flowering, grain filling and maturity.

Planting method, spacing and depth

Plant in rows or basins, approximately 45-60 cm between rows and about 10 cm between plants after thinning. Sow 2-4 cm deep.

Fertiliser and weeding

Apply modest basal fertiliser based on a soil test and top-dress nitrogen only where moisture and yield potential justify it. Weed early, especially during the first 3-4 weeks.

Water and drought

Can produce with roughly 300-500 mm of rainfall and is among the most drought-tolerant cereals. Its short maturity helps it escape late-season drought.

Pests and diseases

Pests include shoot fly, stem borers, aphids, head bugs and Quelea birds. Diseases include downy mildew, ergot, smut, leaf spots and rust. Smut replaces grain with dark powdery masses; ergot produces sticky exudate.

Disease management

Use clean, treated seed and resistant varieties where available, rotate crops, plant early and remove severely diseased plants or residues.

Harvest and post-harvest

Harvest dry, mature heads; dry grain fully before threshing and storage. Protect from insects, rodents and birds.

Soil requirements

Grows on relatively poor sandy or loamy soils and tolerates lower fertility, but performs best on fertile, well-drained soils. It does not tolerate waterlogging.

COWPEAS

Planting period

Plant with effective rains from mid-November to December. Staggered planting may extend fresh-pod harvest where moisture permits.

Planting conditions and land preparation

Use warm, well-drained soil and a moderately fine, weed-free seedbed. Cowpeas can be sole-cropped or intercropped with cereals.

Growth stages

Germination, vegetative growth, flowering, pod formation, seed filling and maturity.

Planting method, spacing and depth

Plant in rows at about 60-75 cm between rows and 10-20 cm within rows, depending on variety. Place seed about 3-5 cm deep.

Fertiliser and weeding

Because cowpeas fix nitrogen, excessive nitrogen is unnecessary. Apply phosphorus and other nutrients based on a soil test; inoculate seed where appropriate. Weed for the first 3-4 weeks until canopy cover develops.

Water and drought

Normally needs around 400-600 mm rainfall. It tolerates drought after establishment, but moisture at flowering and pod filling improves yield.

Pests and diseases

Pests include aphids, thrips, flower bugs, Maruca pod borer, whiteflies and storage bruchids. Diseases include anthracnose, Cercospora leaf spot, rust, bacterial blight and viruses. Viral infection commonly causes mosaic, yellowing and stunting.

Disease management

Use clean or certified seed, resistant varieties, rotation, timely planting and aphid/vector control. Remove heavily infected plants and use labelled products when needed.

Harvest and post-harvest

For dry grain, harvest when most pods are brown and dry; for green pods, pick while full but tender. Dry seed to about 12-13 percent and protect it from bruchids in hermetic storage.

Soil requirements

Prefers well-drained sandy loam to clay-loam soils with pH around 5.5-6.5. It tolerates poorer soils but not waterlogging or salinity.

GROUNDNUTS

Planting period

Plant early with effective rains, generally mid-November to mid-December, to allow adequate time for pod filling.

Planting conditions and land preparation

Use warm, loose, well-drained soil with a fine seedbed. Light soil is important because pegs must enter the ground and pods must develop underground.

Growth stages

Germination, vegetative growth, flowering, pegging, pod development, seed filling and maturity.

Planting method, spacing and depth

Plant in rows roughly 45-60 cm apart and about 10 cm within rows, adjusted for variety. Sow 5-7 cm deep. Use viable, undamaged seed; inoculate where needed.

Fertiliser and weeding

Apply nutrients from a soil test. Nitrogen is normally not required in large amounts because groundnuts fix it. Gypsum at early pegging/around half flowering can supply calcium where soils are deficient. Weed gently and thoroughly for the first 4-6 weeks.

Water and drought

Usually needs about 500-700 mm rainfall. Water is critical at flowering, pegging and pod filling; drought then reduces pod set and kernel size.

Pests and diseases

Pests include aphids, thrips, leaf miners, termites, white grubs and storage pests. Diseases include early and late leaf spot, rust, rosette virus, stem rot and pod rot. Leaf spots cause brown lesions and defoliation; rosette causes severe stunting and yellowing.

Disease management

Use resistant seed, crop rotation of at least 3 years, seed treatment, early planting, field sanitation and fungicides when disease risk warrants. Avoid drought stress and late harvest to reduce aflatoxin risk.

Harvest and post-harvest

Lift when roughly 70-80 percent of pods are mature. Cure/dry pods promptly and reduce pod moisture to about 8-10 percent. Keep dry, clean and ventilated; reject mouldy kernels because of aflatoxin risk.

Soil requirements

Best on loose, well-drained sandy loam or loamy sand, pH about 5.5-6.5. Avoid heavy clay, waterlogged and saline soils.

SUNFLOWERS

Planting period

Plant with effective rains from mid-November to December. It can suit drier zones better than maize once established.

Planting conditions and land preparation

Use a firm, fine, well-drained seedbed in full sun. Ensure early weed control and good seed-to-soil contact.

Growth stages

Germination, vegetative growth, bud formation, flowering, seed filling and maturity.

Planting method, spacing and depth

Plant in rows at roughly 60-75 cm spacing and 20-30 cm within rows. Sow 3-5 cm deep and thin to the desired stand if multiple seeds are placed per station.

Fertiliser and weeding

Apply basal fertiliser following soil-test advice and a moderate nitrogen top dressing where yield potential supports it. Boron deficiency can be important; correct only where confirmed. Weed during the first 4-6 weeks.

Water and drought

Usually needs about 500-700 mm rainfall. It has a deep root system and handles dry periods fairly well, but bud formation, flowering and seed filling are critical.

Pests and diseases

Pests include aphids, cutworms, termites, sunflower moth and birds. Diseases include downy mildew, rust, Alternaria leaf spot and Sclerotinia stem/head rot. Rust causes orange-brown pustules; Alternaria produces dark circular leaf spots.

Disease management

Use certified resistant seed, rotate for several years, avoid waterlogging, manage volunteer hosts and use registered seed treatments or fungicides where appropriate.

Harvest and post-harvest

Harvest when the back of heads turns brown and seeds are hard. Dry seed to roughly 8-10 percent, thresh, clean and store dry and pest-free.

Soil requirements

Prefers fertile, well-drained loams and clay loams, ideally pH 6.0-7.5. Avoid waterlogged or strongly acidic fields.

COTTON

Planting period

Plant during the warm rainy season, commonly mid-November to late December. Earlier planting within the recommended window generally supports better yield potential.

Planting conditions and land preparation

Cotton needs warm soil, full sun and a fine, firm, weed-free seedbed. Prepare drainage and bury or remove old crop residues that may carry pests and diseases.

Growth stages

Germination, vegetative growth, squaring (bud formation), flowering, boll development and boll opening.

Planting method, spacing and depth

Plant in rows about 90-100 cm apart and about 30 cm within the row after thinning. Sow about 3-5 cm deep; retain the recommended plant stand for the variety.

Fertiliser and weeding

Use basal fertiliser at planting and nitrogen top dressing around 6-8 weeks after emergence or early flowering, guided by soil results and expected rainfall. Weed aggressively during the first 6-8 weeks because cotton establishes slowly.

Water and drought

Generally needs about 600-800 mm seasonal water. Flowering and boll filling are the most drought-sensitive stages. Early planting, mulch and moisture conservation help; irrigation is beneficial where available and economical.

Pests and diseases

Key pests are bollworms, aphids, thrips, whiteflies, jassids and red spider mites. Diseases include bacterial blight, Fusarium wilt, Verticillium wilt, boll rot and leaf curl viruses. Wilt causes yellowing, wilting and brown vascular tissue; bacterial blight makes angular dark leaf lesions.

Disease management

Use certified resistant seed, rotation, treated seed, sanitation and regular scouting. Follow an integrated pest-management programme; use only registered pesticides and rotate modes of action to reduce resistance.

Harvest and post-harvest

Pick when bolls open and lint is dry. Pick in several rounds, keep seed cotton clean and dry, and prevent contamination with soil, plastic or coloured fibres before delivery to the ginnery.

Soil requirements

Best on fertile, deep, well-drained clay loams to clay soils with good moisture-holding capacity, usually pH 5.5-7.0. Avoid poorly drained fields.

Reference Notes

- Zimbabwe Department of Agricultural Technical and Extension Services (AGRITEX), Farm Management Handbook: Crop Production.
- Seed Co, Farmers Guide / Agronomy Manual: grain, oilseed and cotton production guidance.
- FAO, Fertilizer-use recommendations and crop production resources for Zimbabwe and Southern Africa.
- FAO, Pests and diseases guide for sorghum and pearl millet.
- UK Government / Natural Resources Institute, Pests and diseases of crops in communal areas of Zimbabwe.
- Use current local pesticide registrations, seed-company labels, and AGRITEX guidance before applying any chemical product.